

# AMIT SINGH RAJAWAT

+91 7999293793

imamitsingh958@gmail.com

in amit-singh

amitsingh-24

Website

## EDUCATION

### Madhav Institute of Technology & Science, Gwalior

B.Tech in Information Technology (Artificial Intelligence & Robotics)

CGPA: 8.65

Aug. 2020 – May 2024

## WORK EXPERIENCE

### Software Engineer

Kloudspot Inc.

June 2024 – Present

Bengaluru, India

- Designed a **Face Recognition Algorithm** in **C++** for **1:1 Verification** and **1:N Identification**, achieving a global rank of 277 in the 1:1 Verification and global rank of 59 in 1:N identification **NIST-Face Recognition Vendor Test (FRVT)**.
- Trained models using **ResNet-200** with **CBAM**, incorporating **knowledge distillation** and **reinforcement learning**, improving accuracy by 13%.
- Developed an **FRS Analysis Tool** using clustering techniques and a **FRS Registration Tool** in **Python**—implementing detection, landmark processing, and recognition for multi-pose registration and real-time tracking with visualizations to monitor passenger movements.
- Developed and maintained scalable **data pipelines** for real-time facial recognition data and integrated **CI/CD practices** for automated deployment and continuous improvement of deep learning models in production environments.

### Software Engineering Intern

Kloudspot Inc.

Feb 2024 – May 2024

Bengaluru, India

- Engineered a high-performance **computer vision inferencing framework** in **C++** using **GStreamer**, **OpenVINO**, and **DLStreamer**, resulting in a **9%** improvement in processing speed.
- Implemented **gRPC** for effective communication between client and server, along with **GStreamer RTSP Server** for seamless real-time video broadcasting which supported high-quality video transmission with minimal packet loss.
- Refined **pre-processing** and **post-processing workflows** to enhance the accuracy and efficiency of model predictions.

### Project Intern (LOR)

Indian Space Research Organisation (ISRO)

Dec 2023 – Jan 2024

Bengaluru, India

- Developed a **vision-based satellite pose estimation system** using the **HRNet algorithm**, improving landmark detection for satellite orientation estimation in varying conditions, achieving 90% accuracy.
- Integrated **deep landmark regression** and **nonlinear pose refinement** techniques, training models on extensive datasets to accurately capture satellite orientations, reducing error rates by 15%.

### Amazon ML Summer School

Amazon

Sept 2023 – Oct 2023

Virtual

- Implemented various **machine learning algorithms** including supervised and unsupervised techniques using advanced **Python libraries**.
- Worked on multiple projects with complex ML models, achieving over **95%** accuracy and leveraging **sequence modeling** where applicable.

## PROJECTS

**StoryCraftAI - Infinite Possibilities in Every Story** | Python, TensorFlow, Keras, NLP, Pandas, Huggingface, CI/CD

(Link)

- Developed** and **optimized** a suite of **deep learning models** for **story generation** using advanced NLP techniques. Improving model accuracies after training **GRU** accuracy increased by **13.46%**, **LSTM** by **33.53%**, **Bidirectional-LSTM** by **25.32%**, and **Bidirectional-GRU** by **9.91%**.
- Integrated** an interactive **Flask-based web interface** for **real-time story generation** and **deployed** the solution via **CI/CD** on **Hugging Face Spaces**.

**GestureSymphony - Control video playback using your hands** | Python, OpenCV, MediaPipe, Pygame, Flask, NumPy, Bootstrap

(Link)

- Engineered a real-time **hand gesture recognition** system using **MediaPipe Hands** and **OpenCV** for detection and control commands, demonstrating strong **computer vision** and **ML** integration skills. Dynamic playlist management to control video playback (pause, play, next, previous).
- Enhanced system efficiency by implementing **lazy loading** and containerizing with **Docker**; established a robust **CI/CD pipeline** for continuous integration and deployment on cloud platforms like **Render** ensuring scalable and reliable real-time performance.

**Snake Game Using RL With Neural Network** | Python, TensorFlow, Keras, Matplotlib, NumPy

(Link)

- Enhanced gameplay** using **Q-learning**, improving the agent's performance through trial and error and state-space representation, and used **boolean features** to improve decision-making and gameplay strategies by 20%.
- Visualized agent performance and evolving strategies with **animated plots** for deeper understanding, improving analysis efficiency by 10%.

Some Other Projects: **Movie Recommendation System with GUI**, **Rainfall Prediction Using ML**, **Text Classification with Tensorflow**

## PUBLICATIONS

- A. Rajawat, A. Manjhi and A. Srivastava, "**Categorization of IRIS Flower using Different Machine Learning Strategies**," [paper]
- A. Manjhi, A. Rajawat and A. Srivastava, "**Design and Analysis of Programmed Face Monitoring System**," [paper]
- Rajawat, A.S., Srivastava, A. "**Recognition of Parkinson's ailment by using various machine learning procedures**". Current Psychology Journal (2024). [paper]
- On Heart Disease** (Communicated in Journal).

## TECHNICAL SKILLS

- Programming Languages:** C/C++ (Proficient), Python (Proficient), JavaScript, HTML/CSS, LaTeX, Shell, MySQL
- Developer Tools and Frameworks:** Git, GitHub, Linux, Docker, VS Code, ONNX, AWS, CI/CD, MLOps
- Libraries:** Pandas, NumPy, PyTorch, TensorFlow, Scikit-learn, Flask, OpenCV, Transformers, MXNet, Keras, LLMs
- Coursework:** DBMS, Machine Learning and Optimization, OS, DSA, CN, Robotics, Computer Vision, NLP, AI

## ACHIEVEMENTS

- Leetcode:** 850+ Problem solved & 1765+ Rating (**Profile Link**), **GFG:** 320+ Problem Solved (**Profile Link**), Featured in **Patrika Newspaper** (Web Link) (Paper Cutting)
- Selected for **IIT Madras Pravartak Undergraduate Fellowship 2023-24** (among 72 in India) (**Link**)